MILITARY SPECIFICATION

GENERATOR SETS, STEAM TURBINE, DIRECT DRIVE, 2000 KW, 450 VOLT, 60 CYCLE A.C.

1. SCOPE

1.1 This specification covers 2000 kilowatt (kw), 450 volt, 60 cycle alternating current (a.c.) steam turbine, direct drive generator sets for Naval shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-T-321 - Thermometers, Self-Indicating, Liquid-in-Glass, for Machinery and Piping Systems.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-V-1137 - Varnish, Electrical-Insulating (for Electro-Motive Equipment).
MIL-R-2729 - Regulator Sets, Voltage, A.C. Generator, Naval Shipboard Use.
MIL-G-3087 - Generator Sets, Steam Turbine (Direct and Alternating Current) (Naval Shipboard Use).
MIL-G-3124 - Generator, Alternating Current, 60-Cycle (Naval Shipboard Use).
MIL-Q-9858 - Quality Control System Requirements.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment.
MIL-T-16049 - Tachometers; Electrical; Self-Generating; Mechanical, Fixed Mounting and Hand Held; and Vibrating Reed.
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having *Rotating Parts (and Associated Repair Parts)*.
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use).
MIL-T-17244 - Thermometer, Self-Indicating, Bimetallic, Shock-Resistant.
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments; Packaging of.
MIL-T-17523 - Turbine, Steam, General Auxiliary (Naval Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-C-18087 - Couplings and Propulsion Clutches for Naval Shipboard Propulsion Units and Auxiliary Turbines.
MIL-C-19836 - Coolers, Fluid, Industrial, Air, Motor and Generator Naval Shipboard.
MIL-T-22051 - Temperature Element, Resistance, Bearings, Babbitt.

STANDARDS

MILITARY

MIL-STD-22 - Welded Joint Designs.
MIL-STD-129 - Marking for Shipment and Storage.

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STANDARDS (Cont'd)

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping Valves, Fittings and Associated Piping Components for Submarine Service.

DRAWINGS

BUREAU OF SHIPS

810-1385860 - Flanges, Bronze, Water Service.
810-1385861 - Flanges NI-Cu, Water Systems.
810-1385873 - Mount, Resilient Type 5M10000, Flanges.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Complete turbine generator sets.-

3.1.1 Each turbine generator set shall be comprised of the turbine and a.c. generator, mounted as an integral unit. Components shall be arranged in the package to occupy a minimum of space and to reduce weight to the lowest practicable figure consistent with other specification requirements. The generator shall be direct connected to the turbine and a reduction gear shall not be used. The turbo-generator is to be mounted on a sound isolated foundation. The foundation is to be specified in the contract or order for the turbo-generator.

3.1.2 The turbine generator sets shall be in accordance with MIL-G-3087 except as otherwise specified hereinafter, and shall have the following classification:

Steam pressure (at throttle) - 285 p.s.i.g. to 750 p.s.i.g.
Steam temperature (at throttle) - Dry and saturated (1.0 percent maximum moisture)
Exhaust pressure - 3 to 7 inches mercury (Hg.) absolute
Parallel operation requirements - See 3.1.10
Rating:
A.c. generator
Kw—2000 kw. continuous (max. rated) (exclusive of excitation)
Voltage—450 volts a.c.
Type—Round rotor
Speed—360 r.p.m.
Class of insulation—Class B or H
Degree of enclosure—Water-air-cooled
Type of voltage regulation and excitation system—Static
Regulators per switchboard—One main
Repair parts—As specified hereinafter.

3.1.3 The turbine generator unit shall operate satisfactorily in accordance with this specification in all respects, including governing, with inlet steam conditions of 285 p.s.i.g. (dry and saturated) to 750 p.s.i.g. (dry and saturated) and with the condenser pressure varying from 3 to 7 inches of Hg. absolute. The turbine shall be designed to withstand an inlet steam pressure of 850 p.s.i.g. The turbine shall be capable of meeting specified operational and life requirements with a maximum of 1.0 percent moisture in the inlet steam.

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3.1.4 The turbine shall be capable of driving the generator at rated capacity, at no guaranteed water rate, when exhausting to a condenser pressure of 10 inches of mercury absolute.

3.1.5 Lube oil coolers, condensers, and air ejectors will not be required.

3.1.6 The overall dimensions and weight of the complete turbine generator set including attached accessories shall not exceed the limits specified (see 6.1).

3.1.7 The generator set shall be capable of withstanding without damage to any components, a 3-phase fully asymmetrical short circuit at the switchboard (assuming 25 feet of interconnecting cable from generator terminals to switchboard) The contractor shall submit a summation of calculations on the anticipated short circuit torque at the generator air gap, the elastic characteristics of all rotors, the shaft loading characteristics, and the short circuit torque multiplying factor. This data shall be accompanied with the manufacturer's statement that all factors have been considered in designing for this short circuit torque condition and that all components will withstand this loading without damage. Data shall be submitted with the installation drawing for review.

3.1.8 The combined unit shall be designed to withstand class HI shock as specified (see 6.1).

3.1.9 The combined unit shall be designed to operate in accordance with the requirements of this specification when subjected to various conditions of trim, list, pitch, and roll as specified in table I.

Table I - Conditions of trim, list, pitch and roll.

	Submerged	Snorkel	Surface
Trim	30 degrees	7 degrees	7 degrees
Pitching	10 degrees	10 degrees	10 degrees
Time of complete pitch cycle	60 seconds	6 seconds	6 seconds
List	15 degrees	15 degrees	15 degrees
Rolling	30 degrees	30 degrees	60 degrees below 10 knots and 30 degrees above 10 knots
Time of complete roll cycle	12 seconds	12 seconds	8 seconds

3.1.9.1 Degrees indicated are measured from vertical to either side for list and roll, and from the horizontal fore and aft plane, up or down for trim and pitch. The conditions of permanent list and roll shall not be considered additive, neither shall conditions of permanent trim and pitch be considered additive. The resultant angle of inclination from the following combinations shall be met:

(a) List and trim.
(b) List and pitch.
(c) Roll and trim (surface only).
(d) Roll and pitch.

The period is the time for one complete cycle, for example, the time required for the ship to pitch or roll from one extreme to the other and return.

3.1.10 The turbine and a.c. generator shall be designed for automatic parallel operation with division of load in accordance with MIL-G-3087, except that the overall speed droop of the units from 0 to 100 percent load may be 4.2 percent, and the necessary governor adjustments and regulator components shall be furnished. The units, however, are not intended for automatic parallel operation but will be required to operate in parallel under manual control for short periods for the purpose of transferring load from one generator to another. The governing and regulating systems shall be so adjusted that operation during such transfer of load shall be stable and controllable at any pressure throughout the pressure range of 285 pounds per square

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inch gage through 750 p.s.i.g. with the exhaust pressure at any value between 3 inches Hg. absolute and 7 inches Hg absolute.

3.1.11 The governing system shall be designed to meet the speed regulation requirements of MIL-G-3087 except that the droop of lines for limits of load-speed curves may be 4.2 percent. Since the units are not intended for automatic parallel operation, however, the governor shall be so adjusted at the factory to provide the minimum possible speed droop at the low pressure point (285 p.s.i.g.) at which the requirements for stable transfer of load (see 3.1.10) can be met.

3.1.12 Resistance temperature elements (RTE) shall be provided for measuring the temperature of the turbine and generator bearings, and the generator stator coils and air. The RTE's used for measuring bearing temperatures shall be embedded in the bearing babbitt. The generator air RTE's shall be located so as to measure the air exhausting into and from the cooler. At least six embedded RTE's shall be provided in the stator windings, the first shall be placed as near as possible to the estimated hot spot of the winding with the remainder spaced at equal intervals of 60 degrees. All RTE's shall be brought out to a common terminal board enclosed with a dripproof box. The external wiring connecting bearing RTE to the common terminal board shall be firmly supported and protected by use of rigid or flexible conduit. Measuring instruments or meters will not be required under this specification. In addition to RTE's, thermometers shall be provided for indication of turbine and generator bearing temperatures.

3.1.13 The combined unit shall comply with the balance and vibration requirements of type II of MIL-STD-167 except that the requirement to record residual unbalance in units of ounce-inches is waived.

3.1.14 In addition to the means used for shop balancing, the turbine and generator shall each have balance rings provided in two planes which can be used for rebalancing in the field. The rings shall provide for balancing by either fastening weights in dovetail slots or by threading weights into tapped holes. The balance rings shall be accessible (with a minimum amount of disassembly) after the units are assembled. Access openings and special tools shall be supplied, as required for making in-place balance corrections to turbine and generator rotors. In making balance corrections in the field, after units are installed, it shall not be necessary to lift turbine casing, remove generator air shields, or perform other major and time-consuming operations. Specific approval by the Bureau of Ships is required for proposed accessibility method to make balance corrections.

3.1.15 Welding and allied processes shall be in accordance with MIL-STD-278 and MIL-STD-22.

3.1.16 All effort shall be made to provide equipment with noise reduced to the minimum degree possible consistent with performance requirements. The following features shall be incorporated to the degree feasible:

(a) Precision machining of rolling and rubbing parts, slip rings, and fitted parts.
(b) Sharp cutoffs and turbulence shall be avoided in the ventilation system. Ventilation ducts and housings shall be damped if necessary, to prevent flow excited vibration.
(c) Static and dynamic balancing to the finest degree attainable.
(d) Most favorable combination of following design features:
    (1) Number of slots per pole pitch.
    (2) Number of slots magnetically under one pole.
    (3) Slot frequency as a function of the natural frequency of the magnetic frame.
    (4) Skewing of stator slots.
    (5) Skewing of pole tips and fairing of pole tip edges.
(e) Spring mounting of stator punchings.
(f) Closer tolerance on air gap symmetry.
(g) Direct drive governor.

3.1.17 Piping, valves, and pipe fittings shall be in accordance with MIL-STD-438.

3.1.18 A turbine generator set subbase and intermediate piping shall be supplied with each ships worth of equipment (two turbine generator sets). The following requirements shall apply:

3.1.18.1 Each turbine generator set subbase shall provide support for two turbine generator sets (one port unit, one starboard unit). The subbase shall be designed for shipboard installation on Government furnished type 5M10000 mounts shown on Drawing 810-1385873. The strength and rigidity of the subbase shall be such that satisfactory operation of turbine generator sets is maintained, under operational conditions including shock

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specified for turbine generators on order. The subbase shall meet this requirement whether supported as intended or solidly supported at each of the four extreme corners.

3.1.18.2 Access shall be provided on the subbase for the use of shipbuilder furnished hydraulic jacks to raise the subbase to permit removal of resilient mounts. (It is not intended to do this at sea but may be done while tied to the dock.) Guide posts shall be provided to aid in the raising of the subbase and shall be suitable for use under a permanent list and trim of 5 degrees. The machinery vendor and the shipbuilder shall work closely together in establishing the procedure for accomplishing the removal of resilient mounts and expansion joints. Intermediate piping shall be supplied with each subbase to extend turbine generator service connections to accessible locations. The steam inlet piping between the anchor point on the subbase and the turbine throttle valve shall have sufficient takedown connections so that it may readily be removed for repairs to the turbine generator set. The location of connections to be made by the shipbuilder shall be as approved by the Bureau of Ships for each ships worth of equipment (one subbase and two turbine generator sets).

3.1.19 Means shall be provided for rotating turbine and generator rotors at slow speed to prevent bowing of the turbine rotor when the turbine generator set is at standby on an operating condenser and with sealing steam being admitted to turbine glands. The leakage of steam into the turbine from gland packing worn to three times design clearance and governor valves steam cut to the point of unstable governor operation shall not cause the device to lose control of turbine speed. The device shall be capable of maintaining a rate of rotation under 300 r.p.m.

3.1.20 The design of all equipment shall be approved by the bureau or agency concerned prior to manufacture.

3.2 Turbines.- The turbines shall be in accordance with MIL-T-17523, except as otherwise specified hereinafter.

3.2.1 For materials subjected to temperatures below 650°F., allowable stresses and appropriate safety factors shall be based solely on the yield strength of the materials.

3.2.2 The emergency power requirements shall not apply.

3.2.3 The turbine manufacturers' shall furnish a drawing showing the application of insulation.

3.2.4 The lubrication and cooling requirements shall apply except that integral lubricating oil piping shall be furnished to receive oil from a ship's turbine generator lubriaction system and deliver the oil to all bearings at not less than 10 p.s.i.g. Therefore, the design and test requirements pertaining to a self-contained system, oil sump, pumps (attached and hand), strainer, relief valve, cooler, and four-way cross cock, shall not apply. The designs shall be based on the use of Navy symbol 2190TEP oil.

3.2.5 The wheels and shaft shall be machined from a single forging.

3.2.6 Special material requirements for turbines. Special material requirements for turbines shall be as follows:

(a) Diaphragms - 12 to 13 percent chrome steel or carbon steel with sealing surfaces and moisture impingement areas protected with an erosion and corrosion resistant material. Inlay materials shall be as approved by the Bureau of Ships. Material in steam path of all diaphrams shall be 12 to 13 percent chrome steel.
(b) Valve seats - Stellited.
(c) Steam admission half of first stage shell shall be protected with 12 to 16 percent chrome.
(d) Turbine casings between diaphragms and all steam sealing surfaces shall be lined with 12 to 16 percent chrome.
(e) Internal moisture collectors shall be provided, where necessary.

3.2.6.1 In lieu of the requirements specified in 3.2.6(c) and (d), consideration shall be given to furnishing a 12 percent chrome casing.

3.2.7 Rotor. The rotor shall be designed so that the average stress in the most highly stressed cross-section of the bladed rotor, under design operating conditions, shall not exceed a stress value corresponding to a minimum factor of safety of 2.0, based upon the following:

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(a) The yield strength (0.02 percent offset) of the material.
(b) Stress rupture of the material.

3.2.8 Blades.- The blades shall be designed so that the average stress in the most highly stressed cross-section of the blades, under design operating conditions, shall not exceed a stress value corresponding to a minimum calculated factor of safety of 2.0, based upon the following:

(a) The yield strength (0.02 percent offset) of the material.
(b) Stress rupture of the material.

3.2.9 The requirements for a gland seal manifold and a compound pressure gage to be used therewith shall not apply. In lieu thereof, connections to the glands shall be provided to connect to self-actuating gland seal regulators which are being furnished separately. The glands and piping shall be designed to withstand a steam pressure of 150 p.s.i.g. in event of failure of the regulator.

3.2.10 A turbine casing relief valve shall not be supplied; a back pressure trip shall be furnished.

3.2.11 The governing system will be supplied with oil from the ship's control oil system at approximately 75 p.s.i.g.

3.2.12 The trip throttle valve requirements shall apply except that, when the trip throttle valve has functioned to trip the turbine, it shall be necessary to reset the actuating devices to their normal operating position and to turn the throttle valve handwheel to the close position before steam may again be admitted to the turbine. The valve shall normally be held open by oil pressure from an external source. The valve seat shall be removable from the valve body. The overspeed trip shall be hydraulically operated, (no mechanical latches or linkages are permitted) and so designed that, when the unit is subjected to shock, this shock shall not cause the trip to close the throttle valve.

3.2.13 Thermometers for measuring turbine and generator bearing temperatures as well as temperatures other than those specified in 3.1.12, shall be in accordance with MIL-T-17244 or type II of GG-T-321.

3.2.14 In addition to provision for the use of a portable hand tachometer, a speed indicating device conforming to type IC/EFC or type IC/EFD of MIL-T-16049 shall be provided. System inaccuracy shall not exceed 1-1/4 percent. A 4- to 4-1/2-inch diameter indicating instrument shall be supplied for mounting on shipbuilder furnished turbine generator set gageboard. The meter shall cover a range of speed from 0 revolutions per minute to 11 percent above rated.

3.2.15 All inlet steam fittings and connections, such as steam chests, throttle valves, and nozzle chambers shall be tested hydrostatically after assembly to a pressure of 1275 p.s.i.g.

3.2.16 The steam inlet connection shall be located on the turbine foundation for connection to a flexible steam joint. The foundation and location of steam inlet is to be specified in the contract or order for the turbo-generator. The flexible steam joint will not be required under this specification.

3.2.17 The exhaust connection shall be circular for use of a flexible steam joint. The flexible steam joint will not be required under this specification.

3.2.18 The gageboard and the pressure gages normally mounted on the gageboard are not required.

3.2.19 The requirements for the overspeed test shall apply except that the maximum reading of the vibration indicator shall not exceed 0.002 inch double amplitude on the shaft on either end of the turbine.

3.2.20 The governor synchronizing motor shall be a single phase, readily reversible, commutatorless a.c. motor in accordance with MIL-M-17059.

3.2.21 The turbine steam rate shall not exceed 18.2 pounds per kilowatt-hour with inlet steam conditions of 285 p.s.i.g. (dry and saturated) and with an exhaust pressure of 7 inch Hg. absolute exhaust pressure, when the unit is delivering a total of 1500 kw.

3.2.22 An automatic switch to trip a circuit breaker when the overspeed trip functions will not be required.

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3.2.23 If a flexible coupling is used between the turbine and generators, it shall conform to MIL-C-18087.

3.2.24 Resistance temperature elements (see 3.1.12) furnished for the bearings shall be located in the loaded half shell between the loadings and 20 degrees past the loadline and shall be in accordance with MIL-T-22051.

3.2.25 Bearing journals and thrust collars shall have a 16 microinch or better surface finish.

3.2.26 Turbine exhaust and condenser inlet flange bolting shall be supplied. This bolting shall be suitable for use with a flexible exhaust joint having a flange thickness of one inch.

3.2.27 Interstage turbine drains shall be collected and routed from the turbine in such a manner to prevent collected moisture from impinging on the condenser tube bundle. A funnel type drain may be supplied in the exhaust trunk for this purpose or an external connection may be located on the turbine casing for an external shipbuilder supplied drain line to the condenser.

3.3 A.C. generators.- A.C. generators shall conform to MIL-G-3124, except as otherwise specified hereinafter:

3.3.1 If wound with class H insulation, the generator efficiencies shall be not less than 88.2 percent at 50 percent load and 91.8 percent at 100 percent load.

3.3.2 Bearings shall be of the sleeve type, forced lubricated from the turbine lubrication system. Sight flow fittings shall be provided at the bearing oil discharge.

3.3.3 The generator air cooler shall be in accordance with class 1 of MIL-C-19836.

3.3.3.1 Salt water flanges shall conform to Drawing 810-1385860 or 810-1385861.

3.3.3.2 The cooler "submergence design pressure" and the hydrostatic test pressure shall be the same as that for condensers under the same contract or order.

3.3.4 Temperature indication equipment for the generator and air cooler shall be as specified in 3.1.12. Alarm contact makers are not required.

3.3.5 The short circuit ratio shall not be less than 0.8. Units shall be electrically stable under all operating conditions.

3.3.6 Generators shall be capable of withstanding, without injury to any part, a three-phase short circuit at the terminals for a period of 30 seconds with the generator under the control of the voltage regulator. The generator shall withstand a single phase short circuit between any two terminals for a period of 30 seconds with the generator under control of the voltage regulator. Total temperatures obtained during these periods shall not exceed safe values.

3.3.7 At rated voltage and frequency and at a power factor of 0.5 or less, each generator shall be capable of delivering 150 percent rated current for 1 minute.

3.3.8 The generator (and associated excitation and regulation system) shall be so designed that during the load application and removal test specified in 4.7, the generator terminal voltage shall neither dip below 80 percent nor rise above 120 percent of initial terminal voltage, and shall return to and remain within plus or minus 3 percent of the final sustained voltage within 2 seconds after removal of the 1.89 per unit impedance load.

3.3.9 The requirements of MIL-G-3124 should be used as a guide in determining the values of generator constants. Any variation in calculated generator reactances differing from present limits by more than 10 percent should be referred to the Bureau or agency concerned for approval prior to finalizing design or starting manufacture of components effected by these values.

3.3.10 The requirement for electric heaters shall apply, except that the heaters shall be suitable for operation on 450-volt, 60-cycle, 3-phase power.

3.3.11 No radio noise limits apply.

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3.3.12 The location and arrangement of the generator main terminals, cooler connections, and static exciter terminals shall be as required by the installation. A mounting pad or flange shall be provided for the attachment of a shipbuilder furnished terminal box for the main terminals and for the exciter terminals if required.

3.3.13 Nonstructural parts, such as covers, terminal boxes, housings, ducts, and similar parts, may be of aluminum or aluminum alloy.

3.3.14 The following shall be applicable for a round rotor design:

(a) Collector rings may be of steel or monel.
(b) Collector ring insulation may be of glass melamine.
(c) Slip ring brushes may be grade D.
(d) Slot wedges may be aluminum or beryllium copper.

3.3.15 Excitation and regulation system.-

3.3.15.1 The excitation and voltage regulation system shall be in accordance with type III of MIL-R-2729. The power rectifier and reactors (if feasible) shall be located in the cooling air stream of the generator, and three rectifiers and reactors shall be readily accessible for inspection, removal, and disassembly or replacement. Load current transformers shall be suitable for switchboard mounting.

3.3.15.1.1 The design shall be such as to meet the requirements of 3.3.8.

3.3.15.2 The generator overload requirement specified in MIL-R-2729 shall apply except that the voltage regulator shall provide necessary excitation control to maintain generator voltage for 1 minute.

3.3.15.3 The generators and regulators shall be designed for automatic parallel operation in accordance with MIL-R-2729 except as follows:

(a) The maximum difference in reactive kva supplied by one generator from that supplied by another generator, based on the continuous kva rating of one machine, shall not exceed 15 percent at loads up to 90 percent. At any load from 90 percent to full load, the division of reactive kva shall be within the limits specified in MIL-R-2729.
(b) The circulating current between generators operating in parallel at no load shall not exceed 7.5 percent of the continuous current rating or one generator at any frequency within the band of plus or minus 5 percent of rated frequency.

3.3.15.4 The units are not intended for automatic parallel operation for this application, and tests to demonstrate compliance with 3.3.15.3 are not required. In lieu thereof, the regulator droop circuits shall be adjusted as necessary to conform to 3.1.10.

3.4 Painting.- The a.c. generator shall be painted as specified hereinafter.

3.4.1 Exterior.- Except for shaft and identification plates, exterior parts shall be painted in accordance with MIL-E-917.

3.4.2 Interior.- Electrical insulation of all types, surfaces in contact with lubricating oil or grease, commutators, collector rings, brushes, brush holders, bearings and bearing surfaces shall not be painted. Also peripheries of armature and rotors and any other rotating part of a machine from which centrifugal force may cause the paint to be thrown on the windings when the machine is operated at rated load and rated ambient temperature shall not be painted; insulation varnish conforming to MIL-V-1137, instead of paint, may be applied to such surfaces. Noncorrosive resisting parts (see MIL-E-917) other than specified herein shall be painted in accordance with MIL-E-917 except that one coat of enamel of any suitable color will be sufficient provided an additional coat is added by the prime mover manufacturer.

3.5 Manuals.- Manuals shall be in accordance with type I of MIL-M-15071 except that an introduction section shall be included. The following equipment shall be covered in the manual:

(a) Steam turbine and any associated accessories.
(b) Generator.
(c) Voltage regulator.

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(d) Manual control means.
(e) Temperature indicating and alarm devices.
(f) Governor motor.
(g) Special devices on equipment.

3.5.1 Manuals shall contain reduced size prints of all drawings approved by the bureau or agency concerned for electric equipment, and a list of all drawings approved by the bureau or agency concerned indicating the manufacturer's and the bureau or agency concerned drawing numbers in parallel columns. Manuals shall be approved by the bureau or agency concerned prior to final printing.

3.5.1.1 If offset printing is used in the publication of the manual, a complete set of negatives shall, after completion of the manual, be furnished to the bureau or agency concerned and shall remain in the property of the Government. Regardless of the method of printing used, one glossy print negative of each halftone illustration included in the manual shall be furnished the bureau or agency concerned.

3.5.2 Two copies of the finally approved manual shall be packed with each turbo-generator set at the time of shipment.

3.6 Drawings.- All drawings shall be master drawings in accordance with MIL-D-963, and shall include information as required by the individual equipment specifications.

3.6.1 Installation drawings.- Installation drawings shall include the following:

(a) All connections, electrical and mechanical weights, overall dimensions, and other data necessary to clearly describe the equipment.
(b) An outline of the identification plate showing all plate markings and the location of the plate.
(c) Piping diagrams, showing recommended connections for oil, water steam and exhaust necessary for proper installation and operation of the unit.
(d) Type numbers, ratings, (volts, amperes, r.p.m., and kw.) and name of manufacturer of all electric equipment.
(e) Bearing and lubrication information including bearing pressure, surface speed, and quantity and temperature of oil supply.

3.6.2 All drawings (unless approved master drawings are on file in the Bureau of Ships) shall be submitted to the bureau or agency concerned via the cognizant Government inspector for approval. The installation drawing shall be submitted for approval within 90 days from the date of contract. Drawings for the turbine, generators, and control equipment shall be submitted for approval within 120 days from the date of the contract or order.

3.7 Onboard repair parts.- Onboard repair parts shall be furnished in accordance with the individual equipment specifications.

3.7.1 Federal stock numbers shall be obtained for all parts in accordance with MIL-P-15137.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality control system.- The supplier shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858 supplemented as specified in 4.1.1.1 and 4.1.1.2.

4.1.1.1 Description of procedures specified in MIL-Q-9858 shall be assembled in manual form, and shall include a flow chart showing the development of the product through the manufacturing cycle. The procedures shall indicate organization and responsibility for control of quality.

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4.1.1.2 The measuring and testing device maintenance and calibration schedule shall be available to the cognizant Government representative at all times.

4.2 Tests shall be conducted on the turbo-generator units in accordance with MIL-G-3087 except that an automatic parallel operation test is not required. Test results on the first generator and combined turbo-generator unit shall be forwarded to the bureau or agency concerned for approval prior to shipment of the units.

4.3 Tests shall be conducted on the turbo-generator units with an inlet steam pressure of 285 p.s.i.g. and exhaust pressure of 7 inches Hg absolute. In addition, sufficient tests shall be conducted at pressure conditions between 285 p.s.i.g. and 750 p.s.i.g. and various other exhaust pressures between 3 inch Hg absolute and 7 inches Hg absolute to determine conformance with performance requirements specified herein. The test agenda shall be approved by the bureau or agency concerned.

4.4 The generator set shall be operated continuously for 50 running hours at full rated kw. No adjustments shall be made on the set during the run. Data shall be recorded as required during the 1 hour continuous load test. The test shall be run with throttle steam at 285 p.s.i.g. and 7 inch Hg absolute exhaust pressure.

4.5 Steam consumption shall be measured at 100, 75, 60, and 25 percent load with steam at 285 p.s.i.g. (dry and saturated) and 7 inch Hg absolute exhaust pressure. Consumption shall be determined by measuring the condensate either by weighing or using water meters calibrated to the satisfaction of the Government inspector.

4.6 The turbine shall be operated at the maximum operating speed, with or without load, for at least 30 minutes in each of the following positions:

(a) Shaft inclined, forward end down 30 degrees.
(b) Shaft inclined, forward end up 30 degrees.
(c) Shaft horizontal, turbine base inclined to left 15 degrees.
(d) Shaft horizontal, turbine base inclined to right 15 degrees.

4.7 The voltage dip determination test of MIL-G-3124 and MIL-R-2729 shall not apply. The test specified in 4.7.1 shall be conducted to determine compliance with 3.3.8 and 3.3.15.1.1.

4.7.1 The generator shall be operating initially with its own exciter and under the control of its own voltage regulator. It shall be carrying a constant impedance load of 67.5 percent rated kva. at rated voltage, frequency, and power factor. A separate balanced 3 phase static load having a per unit impedance of 1.40 at any convenient power factor of 0.4 lagging or less shall be suddenly applied. During the time interval between 0.8 and 1.0 seconds after the application of the separate load, the load shall be suddenly reduced by an amount equivalent to dropping a balanced 3 phase static load having a per unit impedance of 1.89 at any convenient power factor of 0.4 lagging or less.

4.8 Radio noise tests shall not be conducted.

4.9 Periodic test data shall include the field current of the generator at ceiling excitation under single and three phase short circuit conditions with voltage regulator operative and generator field hot.

4.10 The overspeed trip system of the first turbine generator set under a contract shall be tested, with the throttle valve in the wide-open position, by suddenly opening the nozzle control valves while the unit is operating at no load and at rated speed. The rate the nozzle control valves are opened shall not be less than the rate that would be caused by the speed regulating governor suddenly sensing zero speed while the turbine is operating at rated speed. The steam inlet pressure used during the test shall be the maximum design operating steam pressure. The turbine shall not exceed 120 percent of rated speed.

4.11 Instruments, gages, or thermometers containing mercury shall not be used at any time during manufacture or testing of the equipment.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

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5.1.1 Equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged in accordance with level A of MIL-P-16298, MIL-P-17286 and MIL-E-17555, as applicable.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Level A.-

5.2.1.1.1.1 Basic equipment and repair parts.-

5.2.1.1.1.1.1 Electrical.- Electrical components and repair parts shall be preserved and packaged in accordance with MIL-P-16298 and MIL-E-17555, as applicable.

5.2.1.1.1.1.2 Turbines, accessories and instrumentations, repair parts and tools.- The turbines, accessories and instrumentations, repair parts and tools shall be preserved and packaged in accordance with MIL-P-17286.

5.2.1.1.2 Level C.- Preservation and packaging shall be sufficient to afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity for early installation. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2.1.2 Packing.-

5.2.1.2.1 Basic equipment.-

5.2.1.2.1.1 Levels A and B.- The main components of the turbo-generator unit shall be packed for level A or B, as specified (see 6.1), in accordance with MIL-P-16298 or MIL-P-17286.

5.2.1.2.1.1.1 Detached assemblies and detached accessories.- Accessories such as gages, and dials detached to afford physical protection shall be packed in containers specified in 5.2.2.

5.2.1.2.1.2 Level C.- The equipment shall be packed to insure carrier acceptance and safe delivery to destination.

5.2.1.3 Marking.- In addition to any special marking required by the contract or order, marking of the interior packages and shipping containers shall be in accordance with MIL-STD-129.

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5.2.2 Repair parts and tools.- Repair parts and tools shall be packed for level B in snug-fitting wood cleated fiberboard, cleated plywood, nailed wood, wirebound, corrugated or solid fiberboard boxes conforming to PPP-B-591, PPP-B-601 (domestic type), PPP-B-621, PPP-B-585, and PPP-B-636 (class 2), respectively. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds. Stock components shall be packed as specified above for individual weights under 200 pounds. When the gross weight exceeds 200 pounds the components shall be packed in boxes in accordance with PPP-B-621 modified by the addition of skids. Components exceeding a gross weight of 500 pounds shall be packed in sheathed crates conforming to MIL-C-104.

5.2.2.1 Index list.- An index list of repair parts and tools shall be inserted in each repair parts box. The list shall be inserted in the index list support located on the interior side of the box cover or shall be suitably placed inside for quick accessibility. The list shall completely itemize the box content and include stock number, nomenclature and quantity. The list shall be enclosed in a waterproof bag or shall be so treated as to be resistant to water, oil and fading.

5.2.3 Manuals.- Manuals shall be packaged and packed in accordance with MIL-M-15071.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Dimensions and weight of generator set required (see 3.1.6).
(c) Requirements for shock (see 3.1.8).
(d) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

Preparing activity:
Navy - Ships
(Project 6115-N113Sh)

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